



**Building a Trading Platform
In Six Weeks
In an Organisation
That Would Really
Rather We Didn't**

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‘Unprovided with original learning, unformed in the habits of thinking, unskilled in the arts of composition, I resolved to write a book’

— Edward Gibbon

The Credit Crisis

- What?
- How?
- When?
- Who?

Prevention

- Assessing risk
- Controlling risk
- Enforcement of measures

The Client's Problem

- A long way behind its competitors
- History of failure with this project
- Non-obvious benefits

The Challenge

- 6 weeks to meet a hard deadline
- Nobody knew what was needed
- Extreme volatility in requirements
- Nothing is going to work as expected

‘To achieve great things, two things are needed: a plan, and not quite enough time’

— Leonard Bernstein

Why Were We There?



How?

- Risk averse world
- Risk averse client
- Risk Management group
- They want stability

How?

- Risk averse world
- Risk averse client
- Risk Management group
- They want stability
- They're not going to get it!

Let's Change Everything

- Process
 - Absolute minimal scope
 - Frequent incremental releases
 - Business experts always available
- Complete change in technology stack

Why Change Technology?

- The organisation's standard stack sucks!
- Java
- Spring
- Flex!
- Protocol Buffers

What we Suggested

- Scala
- LiftWeb
- Specification by Example
- Lots of test automation



The Cast

- Two very inexperienced Java devs
- One very experienced front office dev
- An enlightened project manager
- Plus four of us (but just me initially)

Introductions

David: If you hire experts then don't restrict their technology choices

James: Yup, I get it. ... err

Pasquale: This is a great opportunity!

Joe: I know what to do, it'll take me 5 mins

Lance: Why can't anyone remember anything?

Andrew: Java is dead to me, I didn't sign up for that

Channing: Yeah, a monad really is just a monoid in the category of endofunctors. What's the problem?

Paul: Meh! Can I type now?

Due Diligence

David: I suppose we should compare a Scala / Lift / ajax / comet app with a Java / Spring / Flex app. Go!

Lance: Really? That's a bit unfair. It's one of me versus three of them.

One hour later

Lance: Done!

Team: What? we haven't written our xml yet!

Lance: I did say it wasn't fair.

David: We're using Scala and Lift.

Scala - First Reactions

‘We’re not smart enough’

Functional Programming

James: This is easy. Errr...

Pasquale: This object is immutable. It just depends on some mutable stuff.

Joe: Have you seen the Guava library?

Lance: I'm not sure how to organise stuff any more

Andrew: F.T.S

Channing: I wish I'd listened to the Haskellers in 1997

Paul: Meh! Can I type now?

Type System

James: This is easy. Errr...

Pasquale: Everything is a typeclass!

Joe: The compiler complained, so I made it an `AnyRef` and used `asInstanceOf` later

Lance: I need to do a lot of reading

Andrew: Is that really all a monad is? Simple!

Channing: That's what I've been telling you!!!

Paul: Meh! This is not the kind of typing that I meant.

LiftWeb

James: This is so much easier than Flex / Proto-Buff / Java

Pasquale: Is this all running in the browser?

Joe: Should we use Play?

Lance: I hate templates! Also, let's use PAC.

Andrew: It's easy!

Channing: The simplest, most flexible web framework I've used.

Paul: Meh! Can I type now?

IDEs

James: It's fine!

Pasquale: !

Joe: I'll use Notepad

Lance: Should we all just use Emacs? And Lisp.

Andrew: I've turned off the compiler

Channing: IntelliJ is better than the others this week. I'll use that.

Paul: Meh! Can I type now?

SBT

James: Let me just... hmmm... ah.... what? Something seems to be broken.

Pasquale:!?!@*

Joe: Apparently you can use Maven with Scala

Lance: NO!

Andrew: NO! I did not sign up for that!

Channing: Could everybody please just read the first couple of pages of documentation?

Paul: Meh! Can I type now?

Akka

James: How is this different from Protocol Buffers?

Pasquale: Await... Await... Await...

Joe: Ah! So we have to use `asInstanceOf` here then?

Lance: We seem to have developed a fully interconnected distributed system. Is that a good thing?

Andrew: Let's distribute lots of stuff.

Channing: Could everybody please just read the first couple of pages of documentation?

Paul: Meh! I prefer object spaces. And ReST. And JSON.
Can I type now?

DSLs

- We were told we were required to audit **EVERYTHING**
- We didn't believe it
- But they insisted
- So I built...

DSLs - TreeLog

github.com/lancewalton/treelog

```
val result: DescribedComputation[Int] =  
  "Adding up" ~< {  
    for {  
      x <- 1 ~> ("x = " + _)  
      y <- 2 ~> ("y = " + _)  
      z <- (x + y) ~> ("z = " + _)  
    } yield z
```

Adding up

x = 1

y = 2

z = 3

DSLs

Lance: *for { result ~> “I have done a thing” } yield result*

James: What does ~> mean?

Pasquale: What does ‘for’ mean?

Joe: Why don’t we just use Lift’s built in logging everywhere?

Andrew: Cool. Let’s use that everywhere.

Channing: Excellent. Publish it.

Paul: Meh! I hate logging and auditing. Can I type now?

The Thrilling Conclusion

- We did it!
- We *weren't* included on the LCH's press release
- Our product became a massive sales tool

Two Years Later

- Delivering good stuff every week or two
- Extremely happy business users and sales people
- One defect in production in two years!

Scala - Now

James: It's really cool

Pasquale: This has been a great opportunity!

Joe: It's ok.

Lance: I suppose I'd better learn Scala

Andrew: I'm never writing Java again

Channing: Hmm, Haskell!

Paul: Meh! Can I type now?

Summary

- Biggest benefit came from better engagement with the business
- Technology helped too
- New technology is not to be avoided
- Nobody wants to go back to Java
 - Very happy with Scala



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